

Forensic Ancestry Benchmark: Machine Learning Classification Using Five Ancestry-Informative SNPs from the 1000 Genomes Project.

Abstract

Forensic ancestry inference is increasingly used to generate investigative leads when conventional DNA profiling cannot identify an unknown individual. The present study evaluates the performance of a minimal panel of five ancestry-informative single nucleotide polymorphisms (AISNPs) using data from the 1000 Genomes Project Phase 3 reference dataset.

Genotypes from 2,504 individuals representing five continental superpopulations (AFR, AMR, EAS, EUR, and SAS) were analyzed. Population-specific genotype frequencies, principal component analysis (PCA), machine learning classification, feature importance analysis, and SNP dropout experiments were performed to assess the ancestry-informative value of the selected markers.

Principal component analysis demonstrated clear continental population structure using only five AISNPs. Four machine learning algorithms were evaluated for ancestry prediction, including Support Vector Machine (SVM), Logistic Regression, Random Forest, and Decision Tree classifiers. SVM achieved the highest classification accuracy (91.2%), followed by Logistic Regression (90.8%), Random Forest (90.6%), and Decision Tree (90.2%).

Feature importance analysis identified rs2814778 as the most informative marker. SNP dropout experiments revealed a gradual decline in classification performance as markers were removed, demonstrating the robustness of highly informative AISNPs under simulated degraded DNA conditions.

These findings demonstrate that a compact panel of five carefully selected AISNPs captures substantial continental population structure and supports robust ancestry inference within a forensic genetics framework.

Keywords: forensic genetics, ancestry inference, AISNPs, machine learning, 1000 Genomes Project, population genetics

1. Introduction

Forensic DNA analysis traditionally relies on short tandem repeat (STR) profiling for human identification. However, conventional STR-based approaches may fail when no reference sample or database match is available. In such situations, forensic ancestry inference can provide valuable investigative information regarding the probable biogeographical origin of an unknown individual.

Ancestry-informative single nucleotide polymorphisms (AISNPs) have emerged as powerful markers for ancestry prediction because they exhibit substantial allele-frequency differences among human populations. Previous studies have demonstrated that carefully selected AISNP panels can accurately distinguish major continental population groups and provide useful investigative leads in forensic casework.

The development of large public reference datasets, including the 1000 Genomes Project, has enabled systematic evaluation of ancestry inference methods across globally diverse populations. At the same time, machine learning approaches have increasingly been applied to forensic genetics, providing flexible frameworks for ancestry classification and prediction.

The present study investigates whether a minimal panel of only five AISNPs can capture meaningful continental population structure and support accurate ancestry classification. Using publicly available 1000 Genomes Project data, we benchmark several machine learning algorithms and evaluate classification robustness under simulated SNP dropout conditions.

2. Materials and Methods

2.1 Reference Dataset

Genotype and population metadata were obtained from the 1000 Genomes Project Phase 3 reference dataset. A total of 2,504 individuals representing five continental superpopulations were included:

- AFR (African)

- AMR (Admixed American)
- EAS (East Asian)
- EUR (European)
- SAS (South Asian)

2.2 AISNP Panel

Five ancestry-informative SNPs were selected based on their established utility in forensic genetics:

SNP	Gene / Region
rs2814778	ACKR1
rs3827760	EDAR
rs1426654	SLC24A5
rs16891982	SLC45A2
rs12913832	HERC2/OCA2

2.3 Population Genetic Analysis

Population-specific genotype frequencies were calculated for all markers. Principal Component Analysis (PCA) was performed using numerically encoded genotypes to evaluate population structure captured by the AISNP panel.

2.4 Machine Learning Classification

Four supervised classification algorithms were evaluated:

- Support Vector Machine (SVM)
- Logistic Regression
- Random Forest
- Decision Tree

Models were trained to predict continental ancestry labels and evaluated using held-out test data.

2.5 Feature Importance

Random Forest feature importance scores were calculated to estimate the relative contribution of individual AISNPs to ancestry prediction performance.

2.6 SNP Dropout Analysis

To simulate degraded forensic DNA profiles, progressive SNP dropout experiments were conducted by sequentially removing AISNP markers and evaluating changes in classification accuracy.

3. Results

3.1 Population Differentiation

All five AISNPs exhibited substantial genotype frequency differences among continental populations. Marker-specific enrichment patterns were consistent with previously reported ancestry associations.

3.2 Principal Component Analysis

Principal component analysis revealed clear clustering of continental populations. The first principal component explained 52.7% of total variance, while the second explained 27.6%, together accounting for 80.3% of observed variation.

African populations formed a distinct cluster, whereas East Asian and European populations occupied separate regions of PCA space. Admixed American populations displayed broader dispersion consistent with mixed ancestry.

3.3 Machine Learning Performance

Classification accuracies were:

Model	Accuracy
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Support Vector Machine	91.2%
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Logistic Regression	90.8%
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Random Forest	90.6%
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Decision Tree	90.2%
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The Support Vector Machine achieved the highest overall classification accuracy.

3.4 Feature Importance

Random Forest feature importance analysis identified rs2814778 as the most informative ancestry marker.

SNP	Importance
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rs2814778	0.315
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rs16891982	0.226
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rs3827760	0.216
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rs1426654	0.172
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rs12913832	0.072
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3.5 SNP Dropout Benchmark

Classification performance decreased progressively as markers were removed from the panel. Despite the reduction in available information, ancestry prediction remained relatively robust under moderate SNP dropout conditions.

4. Discussion

The present benchmark demonstrates that a compact panel of five AISNPs captures substantial continental population structure within the 1000 Genomes reference

dataset. The observed classification accuracies suggest that carefully selected ancestry markers can provide informative ancestry predictions despite the small panel size.

The superiority of the Support Vector Machine classifier indicates that ancestry differentiation captured by the selected markers is highly separable within the underlying feature space. The feature importance rankings are consistent with established forensic genetics literature and highlight the importance of rs2814778 for distinguishing African ancestry.

Although the benchmark achieved strong classification performance, several limitations should be acknowledged. The study focused exclusively on continental ancestry inference and did not evaluate fine-scale population structure. Furthermore, only five AISNPs were included, whereas larger forensic panels typically contain dozens of markers.

Future work will expand the benchmark to larger AISNP panels, including the Kidd 55 marker set, and evaluate additional machine learning approaches.

5. Conclusions

A panel of five ancestry-informative SNPs successfully reproduced known continental population structure within the 1000 Genomes Project Phase 3 dataset.

Machine learning classifiers achieved classification accuracies exceeding 90%, with Support Vector Machine providing the best overall performance. Principal component analysis, genotype frequency distributions, and feature importance analyses consistently demonstrated substantial ancestry information contained within the selected markers.

These findings support the utility of compact AISNP panels for forensic ancestry inference and provide a reproducible benchmark framework for future studies.

Data and Code Availability

GitHub repository:

<https://github.com/ag48665/forensic-ancestry-benchmark>

Zenodo archive:

<https://doi.org/10.5281/zenodo.20643550>

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